

### TOPIC 8: CLIMATE AND ENVIRONMENTAL CHANGES

#### 1. Meaning and Causes of Climate Change

Climate change refers to long-term changes in temperature, rainfall and weather patterns. A major driver today is the increase in greenhouse gases (carbon dioxide, methane) from burning fossil fuels, deforestation and some agricultural practices, leading to global warming.

#### 2. Effects of Global Warming

Rising temperatures, melting ice, rising sea levels, more extreme weather (floods, droughts), threats to crops, water supply and health, and loss of habitats for plants and animals.

#### 3. Mitigation and Adaptation

**Mitigation** – reduce greenhouse gas emissions: use renewable energy, save energy, plant trees, reduce deforestation. **Adaptation** – adjust to changes already happening: better water management, drought-resistant crops, flood defences, early warning systems.

#### 4. School-Based Actions

Tree planting, environmental clubs, reducing waste and energy use at school, and learning about climate so that students can act as informed citizens.

#### 5. Summary

Climate change is driven largely by human activities and has serious effects. Mitigation and adaptation, including local actions like tree planting, help protect the environment and our future.

End of Topic 8 – Climate and Environmental Changes

— End of JSS 2 Intermediate Science First Term Notes —